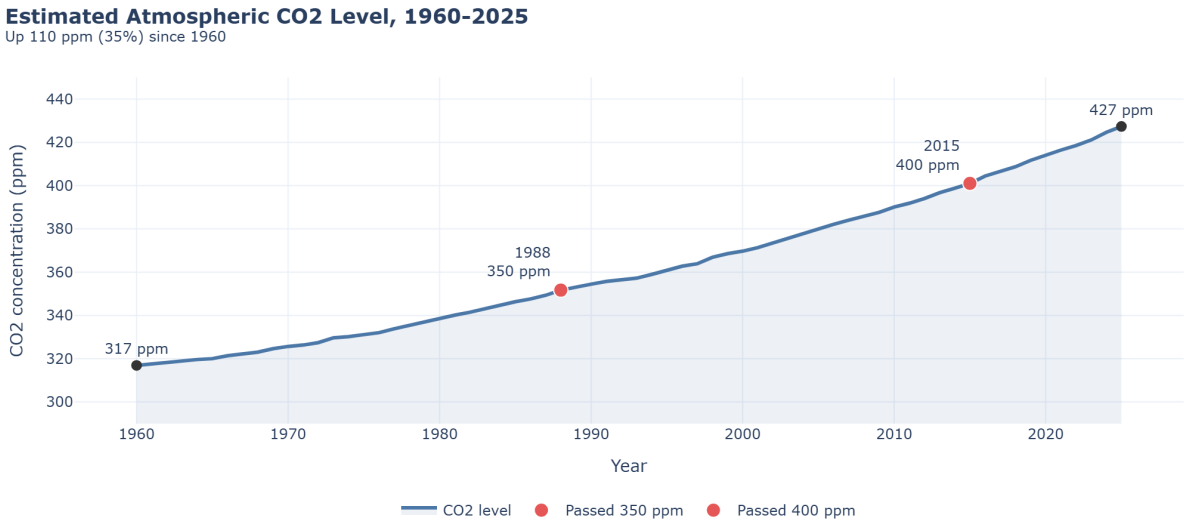
Climate, Emissions and Air Quality Analysis

Questions, graphs and findings from all topics
Generated 15 October 2026

Global Trends

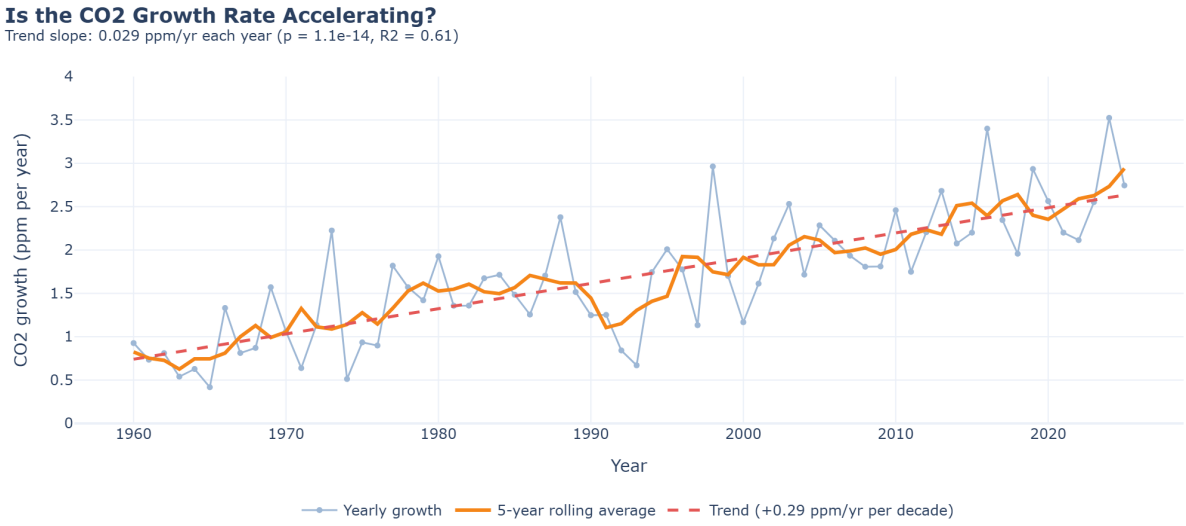
Q1. How has CO_2 changed from 1960 to now?



SUMMARY

Atmospheric CO_2 rose from about 316.9 ppm in 1960 to about 427.4 ppm in 2025, an increase of roughly 110.4 ppm (about 35%). The rise has been continuous: no year shows a decline.

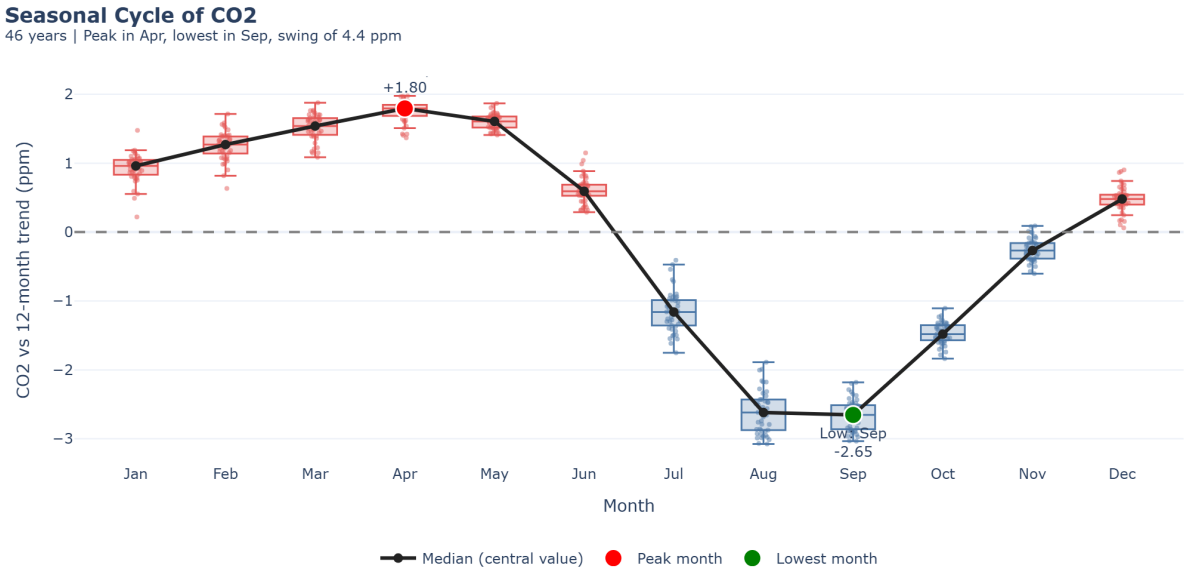
Q2. Is the growth rate of CO_2 itself accelerating?



SUMMARY

Yes: the average annual growth rate of atmospheric CO_2 moved from 0.86 ppm/yr in the 1960s to 2.62 ppm/yr in the 2020s (about 3.0 times higher). A linear trend changes the growth rate by 0.029 ppm/yr every year (about 0.29 ppm/yr per decade, R2 = 0.61). Short-term spikes and dips are mainly driven by El Nino and volcanic events (for example the 1991 Pinatubo eruption) on top of this long-term trend.

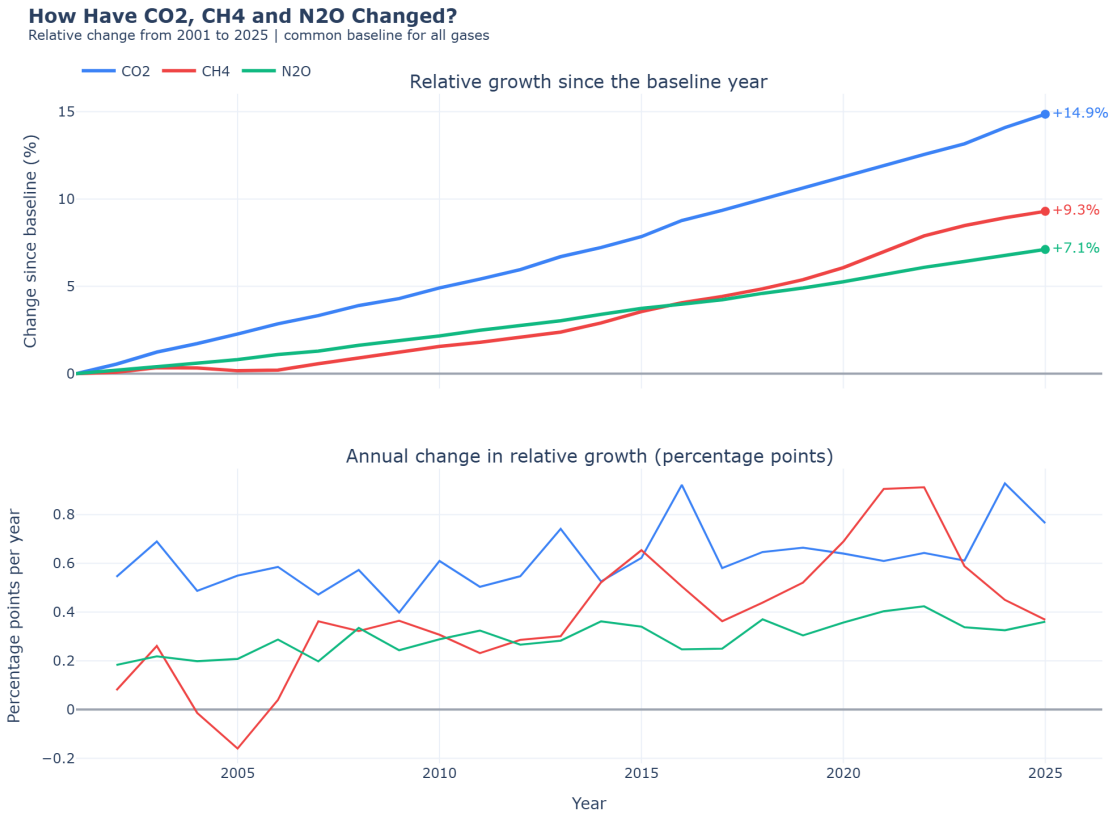
Q3. What is the seasonal cycle of CO_2 and which month peaks?



SUMMARY

Atmospheric CO_2 follows a regular seasonal cycle. After removing the long-term trend with a 12-month moving average, the median across 46 complete years shows CO_2 peaking in Apr (+1.80 ppm above trend), with May close behind (+1.61 ppm). It then falls through the summer to its lowest point in Sep (-2.65 ppm), with Aug almost equal (-2.62 ppm). The seasonal swing is about 4.45 ppm from peak to trough. The cycle arises because plants absorb CO_2 during the summer growing season and release it through decay in autumn and winter. This swing is small compared with the long-term rise of 110 ppm since 1960.

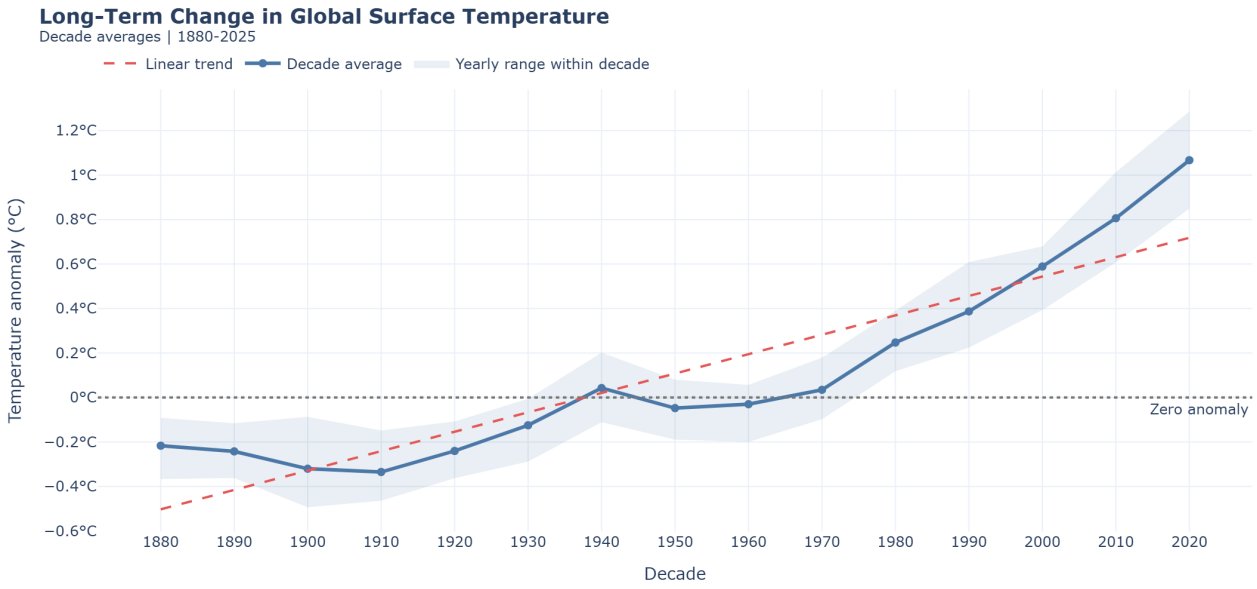
Q4. How have CH_4 and N_2O changed, and are they growing faster or slower than CO_2 in relative terms?



SUMMARY

From 2001 to 2025, atmospheric concentrations of all three greenhouse gases increased, and CO_2 showed the highest relative growth. CO_2 rose 14.86% overall, compared with 9.30% for CH_4 and 7.11% for N_2O. On an annualized basis, CO_2 grew at 0.579% per year, CH_4 at 0.371% and N_2O at 0.287%. In percentage terms, CH_4 accumulated slower than CO_2 and N_2O accumulated slower than CO_2.

Q5. How did global temperature anomaly change by decade, and which decade jumped most?

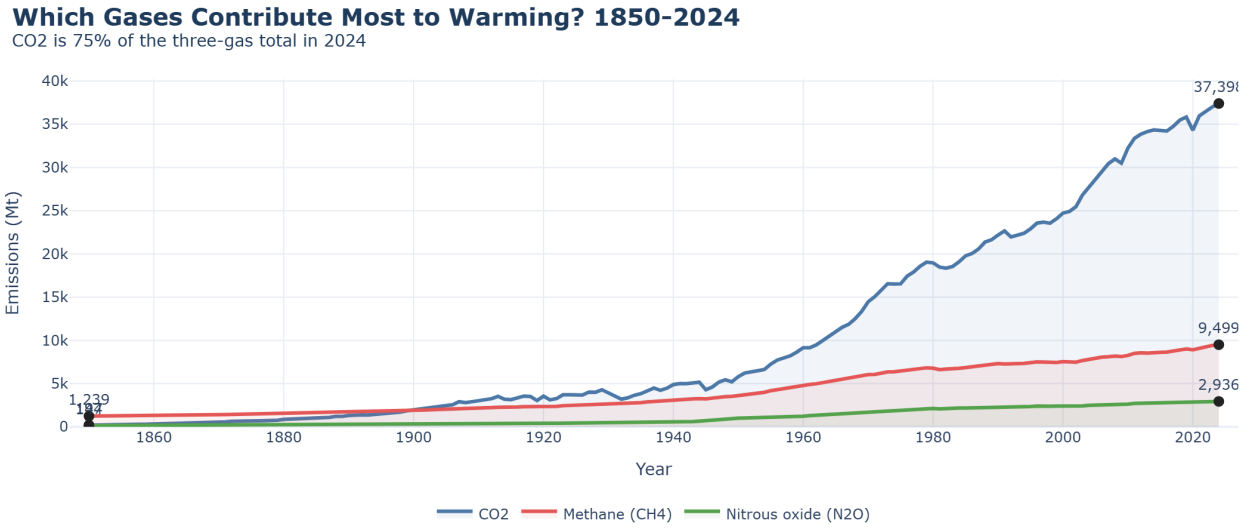


SUMMARY

Global temperature anomalies show a clear long-term warming trend, rising from an average of -0.217 °C in the 1880s to +1.067 °C in the 2020s, a difference of 1.284 °C between the two decade averages. The largest increase between consecutive decade averages occurred from the 2010s to the 2020s, rising by 0.261 °C. Note that the 2020s cover only 6 years so far, so that decade is incomplete. The linear trend across decade averages is +0.087 °C per decade.

Which Gases Contribute to Warming

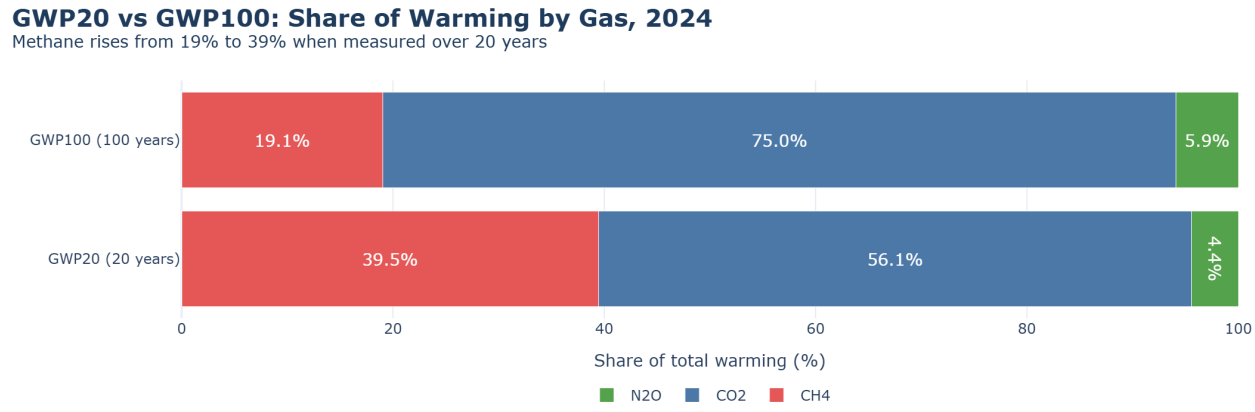
Q1. Which gases contribute most to warming?



SUMMARY

CO_2 is the largest contributor to warming. In 2024 it accounts for about 75.0% of the combined three-gas total (CO_2, methane and nitrous oxide). Methane (CH_4) is second at 19.1%, and Nitrous oxide (N_2O) is third at 5.9%. Together, the two non-CO_2 gases contribute about 25% of the total, so cutting CO_2 matters most, but methane is the second lever. These shares assume the methane and N_2O columns are already in CO_2-equivalent using the dataset's GWP100 factors.

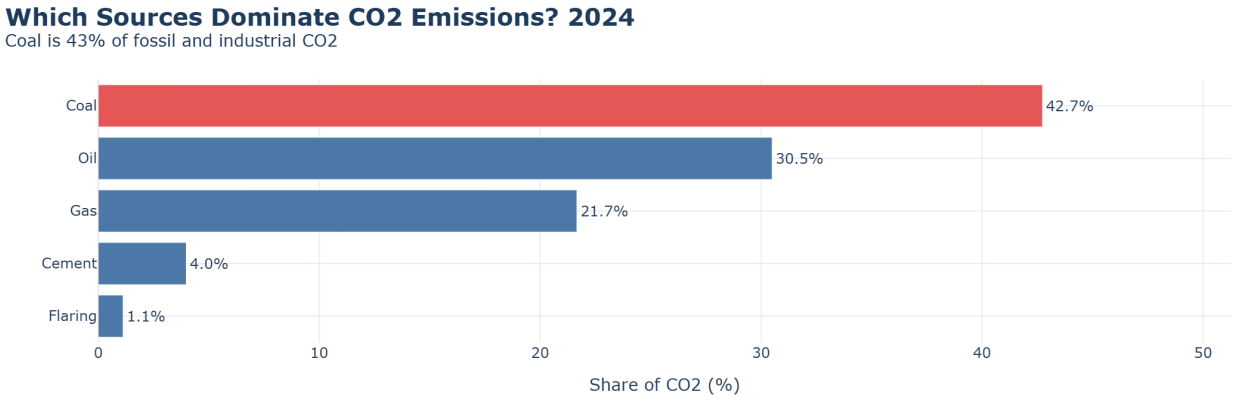
Q2. How does the picture change if you compare over 20 years instead of 100 years (GWP20 vs GWP100)?



SUMMARY

The ranking stays the same (CO_2, CH_4, N_2O), but the gap narrows sharply. Methane has a GWP of 82.5 over 20 years against 29.8 over 100, because it traps heat strongly but breaks down in about a decade. Using 2024 emissions as the baseline, methane's contribution rises from about 19% to 39%, while CO_2 falls from 75% to 56%. N_2O moves from 5.9% to 4.4%, because the total grows when the short-lived gas is weighted more.

Q3. Which emission sources dominate CO_2 gas?

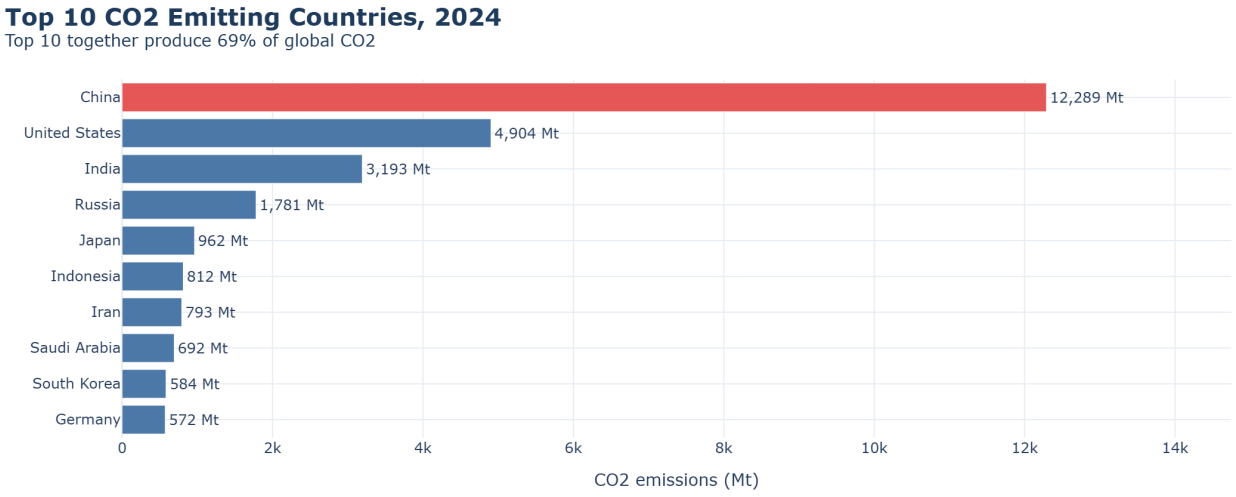


SUMMARY

Coal and Oil dominate, together producing about 73% of fossil and industrial CO_2 in 2024. Coal is the largest single source at 42.7% (15,800 Mt), compared with 35.6% in 1974. Gas gained the most share over the 50 years (13.7% to 21.7%), while Oil lost the most (46.3% to 30.5%). Cement (4.0%) and Flaring (1.1%) are minor sources. The mix has shifted between fuels, which means power generation and industry as well as transport drive the largest shares of CO_2.

Countries and Emissions

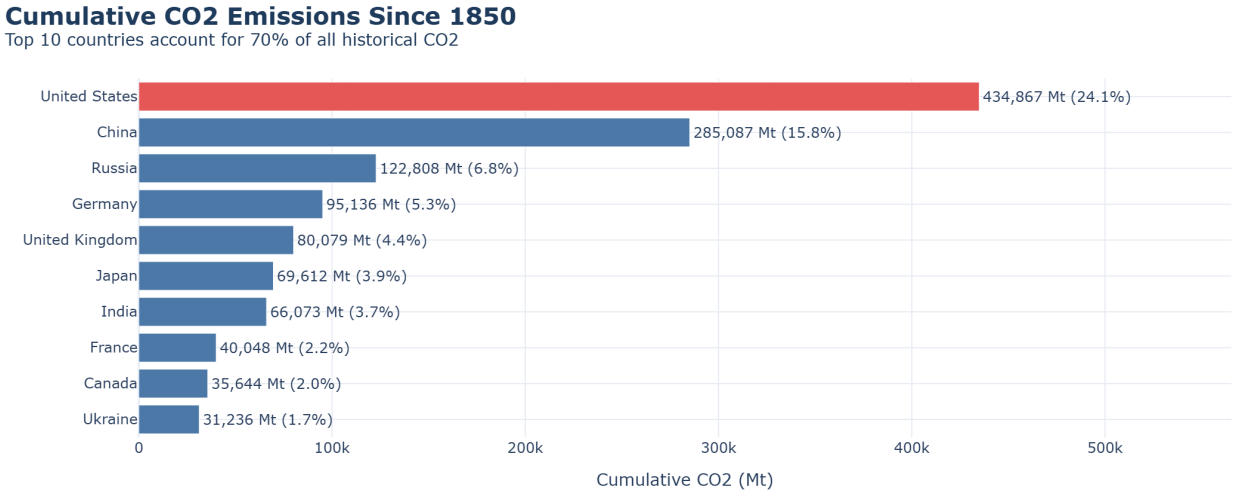
Q1. Which countries emit the most today?



SUMMARY

Emissions are highly concentrated. China emits 12,289 Mt, or 31.8% of the global total. United States is second at 4,904 Mt (12.7%) and India third at 3,193 Mt (8.3%). Together these three emit about 53% of world CO_2. The top 10 countries produce 68.9%, so action by a small group of countries has an outsized effect on global totals.

Q2. Which countries contributed the most cumulatively since 1850?

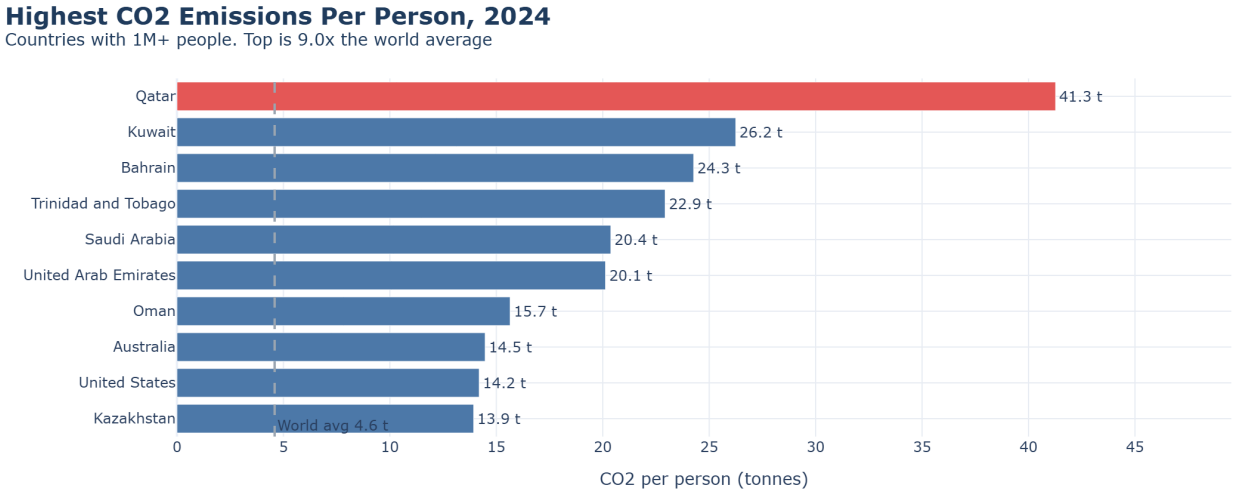


SUMMARY

History tells a different story from today. United States leads with 434,867 Mt (24.1% of all historical CO_2), ahead of China at 285,087 Mt (15.8%). Russia (6.8%), Germany (5.3%) and United Kingdom (4.4%) follow, which shows the lasting weight of early industrializers. Among today's three largest emitters, China ranks #2 (15.8%), United States ranks #1 (24.1%), India ranks #7 (3.7%) historically. The top 10 account for 69.9% of all cumulative emissions, and United States carries roughly 1.5 times

the historical emissions of China.

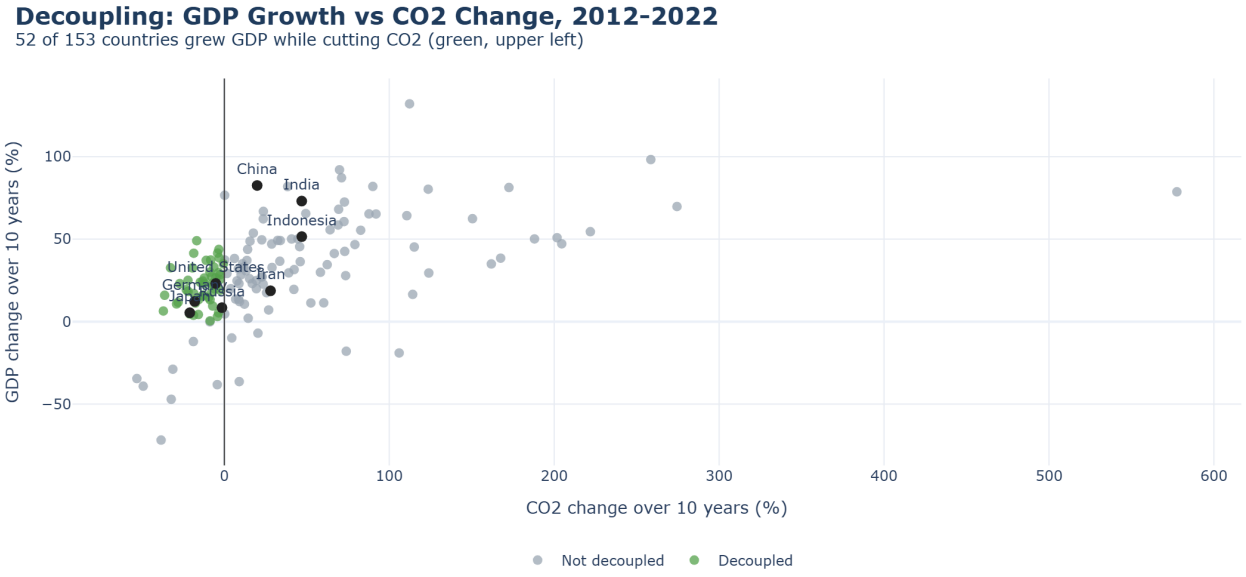
Q3. Who emits the most per person?



SUMMARY

The highest per-person emitters among countries with 1M+ people are Qatar (41.3 t), Kuwait (26.2 t), Bahrain (24.3 t), Trinidad and Tobago (22.9 t), Saudi Arabia (20.4 t), United Arab Emirates (20.1 t). The leader, Qatar, emits 9.0 times the world average of 4.59 tonnes per person. Among the six largest total emitters, per-person levels are: China 8.7 t, United States 14.2 t, India 2.2 t, Russia 12.3 t, Japan 7.8 t, Indonesia 2.9 t. Total and per-person rankings therefore tell very different stories: a country can dominate world totals while its per-person level is moderate, or the reverse.

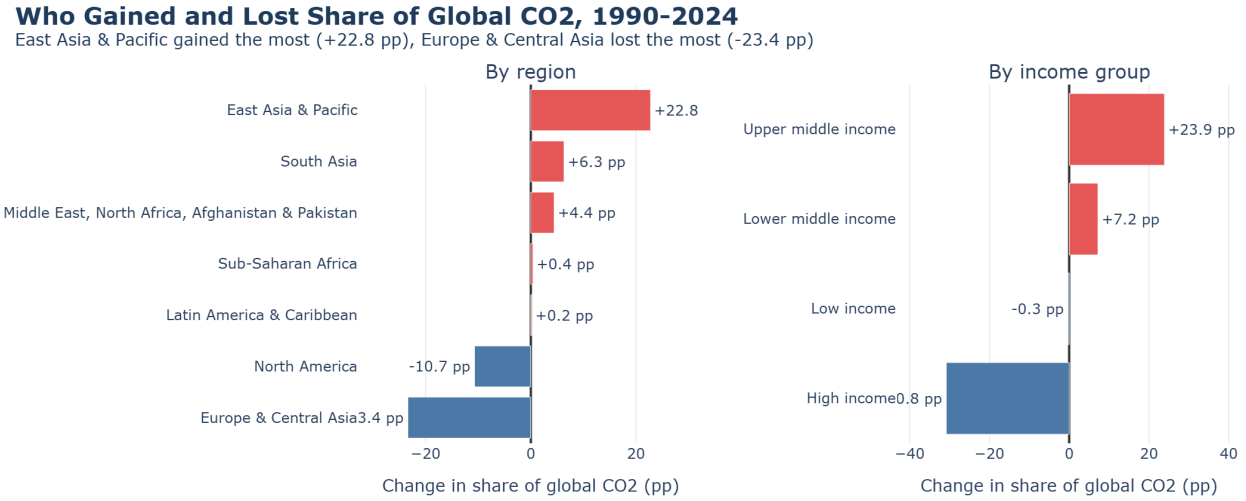
Q4. Which countries decoupled emissions from GDP growth?



SUMMARY

Decoupling is real but not universal: 52 of 153 countries (about 34%) grew GDP while cutting CO_2 between 2012 and 2022. Large emitters on the list include United States (GDP +23.1%, CO_2 -5.2%), Russia (GDP +8.5%, CO_2 -1.4%), Japan (GDP +5.4%, CO_2 -20.9%), Germany (GDP +12.3%, CO_2 -18.0%), South Korea (GDP +29.7%, CO_2 -3.4%). The strongest cuts came from Greece (GDP +6.5%, CO_2 -37.0%), United Kingdom (GDP +16.0%, CO_2 -36.2%), Estonia (GDP +32.6%, CO_2 -32.6%), Finland (GDP +10.8%, CO_2 -29.0%). Among the largest emitters that did not decouple are China (GDP +82.5%, CO_2 +19.9%), India (GDP +73.1%, CO_2 +46.9%), Iran (GDP +18.7%, CO_2 +28.0%), Indonesia (GDP +51.6%, CO_2 +47.0%). Some decouplers cut emissions only marginally, so a decoupled flag does not always mean meaningful reductions.

Q5. Which regions and income groups changed their share most over 1990 to now?

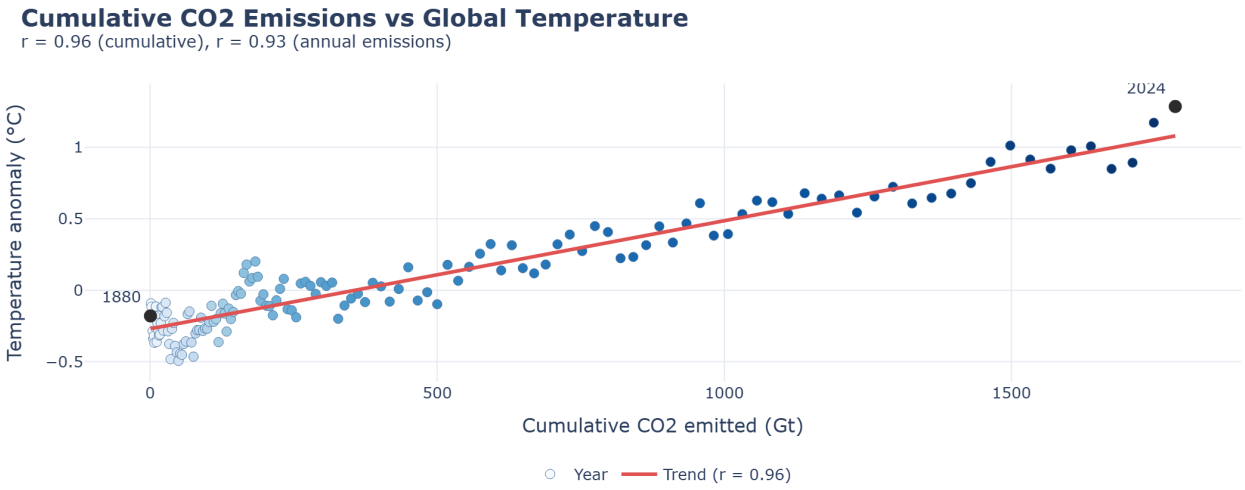


SUMMARY

The centre of gravity of global emissions has shifted between 1990 and 2024. By region, the biggest gains were East Asia & Pacific (21.6% to 44.4%, +22.8 pp), South Asia (2.7% to 9.0%, +6.3 pp), Middle East, North Africa, Afghanistan & Pakistan (4.4% to 8.8%, +4.4 pp); the biggest losses were Europe & Central Asia (39.4% to 16.1%, -23.4 pp), North America (25.3% to 14.6%, -10.7 pp). By income group, the gains were Upper middle income (26.5% to 50.4%, +23.9 pp), Lower middle income (5.2% to 12.4%, +7.2 pp), while the losses were High income (67.5% to 36.6%, -30.8 pp), Low income (0.9% to 0.6%, -0.3 pp).

Emissions and Temperature

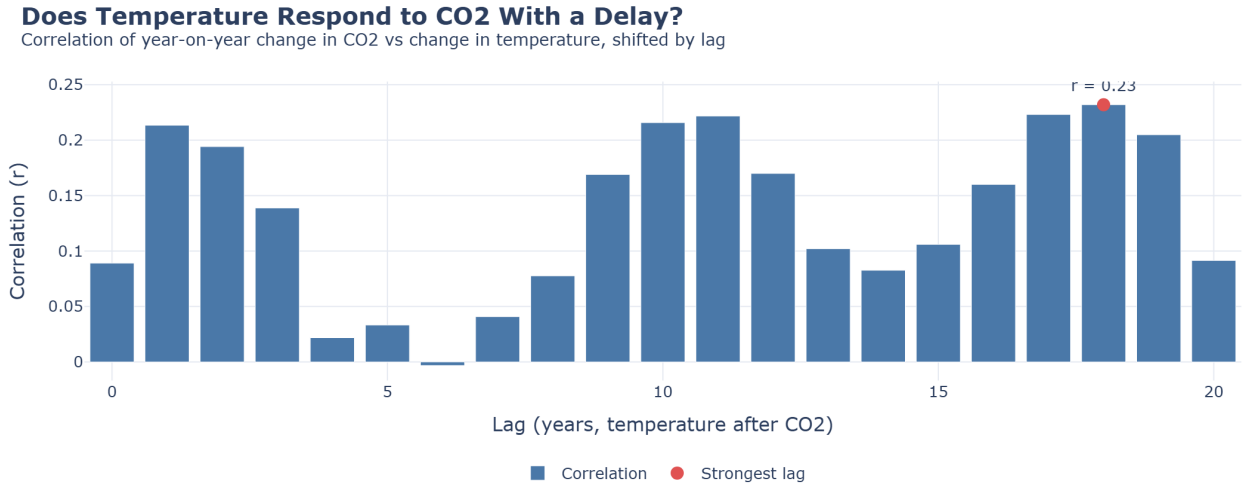
Q1. How strongly do emissions correlate with temperature?



SUMMARY

Over 1880-2024, temperature correlates strongly with cumulative emissions (r = 0.96) and also with annual emissions (r = 0.93). Each additional 1,000 Gt of CO_2 is associated with about 0.76 °C of warming. Both series trend upward, so correlation alone overstates certainty, but the cumulative relationship matches the physical expectation that warming tracks total CO_2 added.

Q2. Does temperature respond with a delay to CO_2?

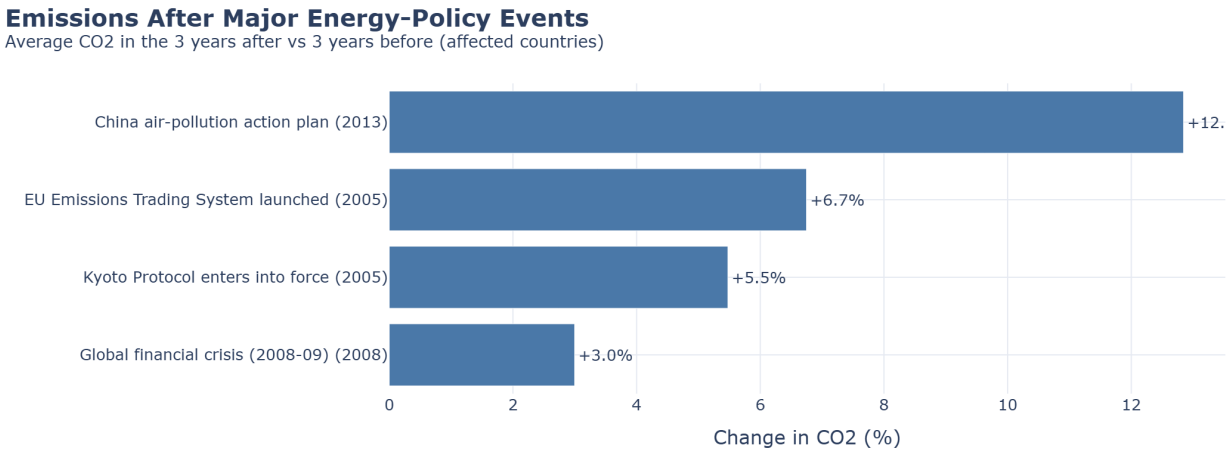


SUMMARY

Using smoothed year-to-year changes, the strongest correlation appears at a lag of 18 years (r = 0.23); at zero lag r = 0.09. This suggests temperature lags emissions changes. Annual temperature is noisy (El Nino, volcanoes), so treat the exact lag as indicative rather than precise.

Policy Events and Emissions

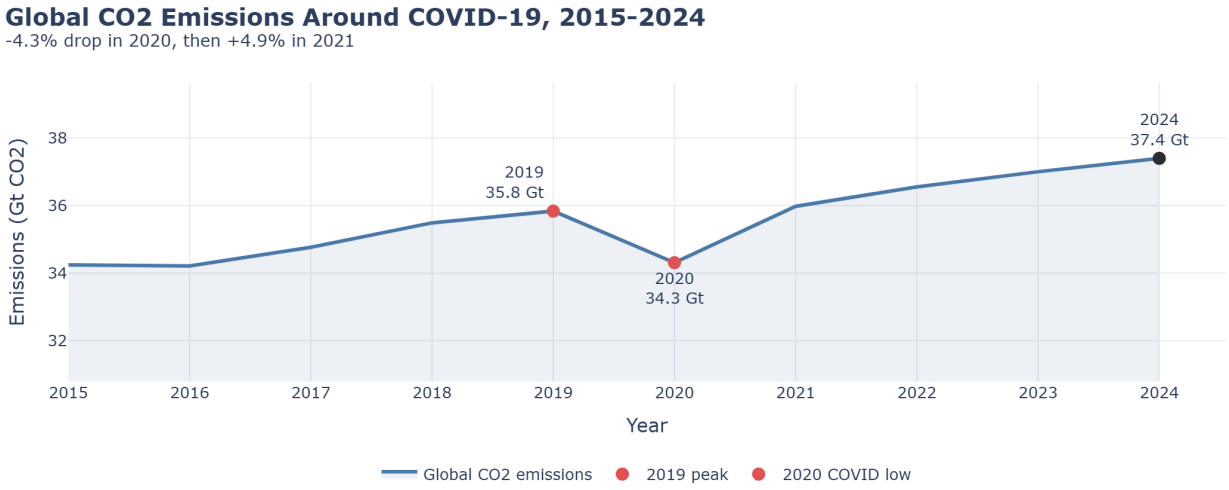
Q1. What happens to emissions after major energy-policy changes?



SUMMARY

Of 4 events analysed, 0 were followed by lower average emissions in the countries involved. The largest decline followed "Global financial crisis (2008-09)" (2008, 3.0%), while "China air-pollution action plan" (2013) saw a +12.8% change. This is a before/after comparison, not proof of causation, since the wider economy and other policies move at the same time.

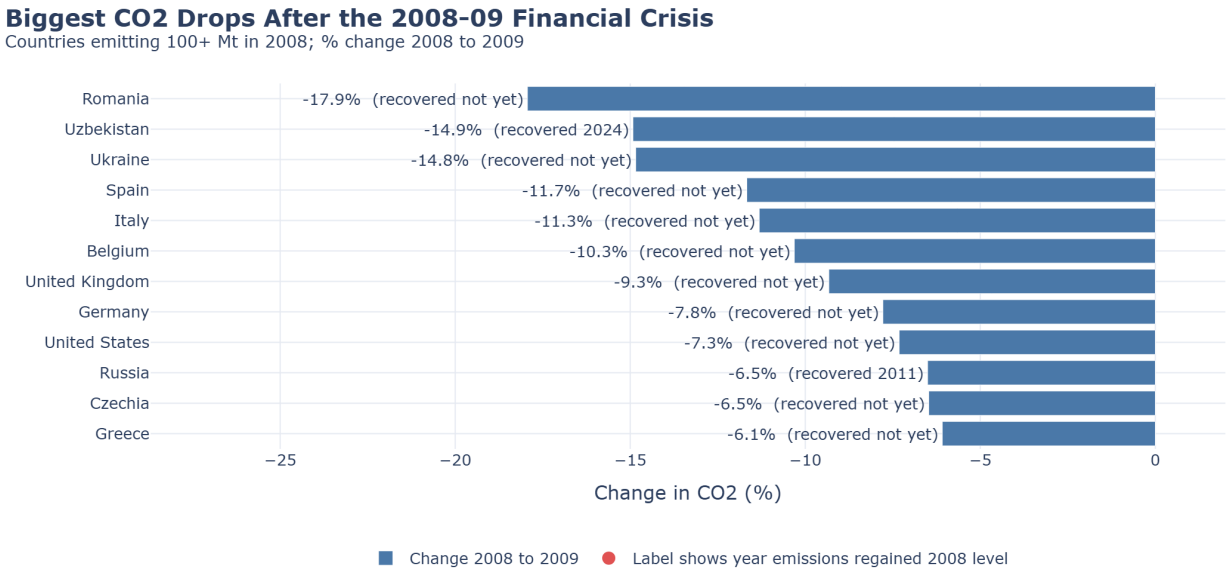
Q2. Did the 2020 COVID drop in CO_2 show up, how large was it, and how quickly did emissions rebound?



SUMMARY

Global CO_2 emissions fell from 35.84 Gt in 2019 to 34.31 Gt in 2020, a drop of 1.53 Gt (-4.3%), the sharpest single-year fall in the recent record. Emissions rebounded 4.9% in 2021 and returned to the 2019 level in 2021. The drop was therefore real but temporary: it reflected lockdown-related demand, not structural change.

Q3. Which countries saw the biggest CO_2 drop after 2008-09 and recovered fastest?

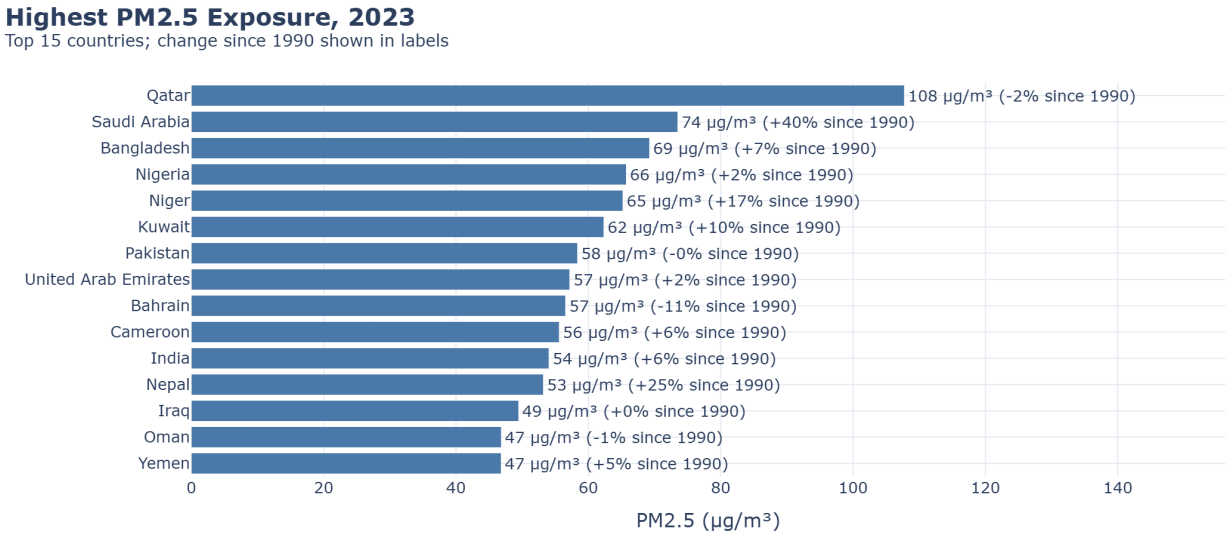


SUMMARY

Among large emitters, the steepest falls were Romania (-17.9%), Uzbekistan (-14.9%), Ukraine (-14.8%). The fastest to regain their 2008 emissions were Russia (2011), Uzbekistan (2024). Countries with the deepest drops did not always recover fastest; recovery depended on whether the fall was cyclical (demand) or structural (fuel switching, deindustrialisation).

Air Pollution

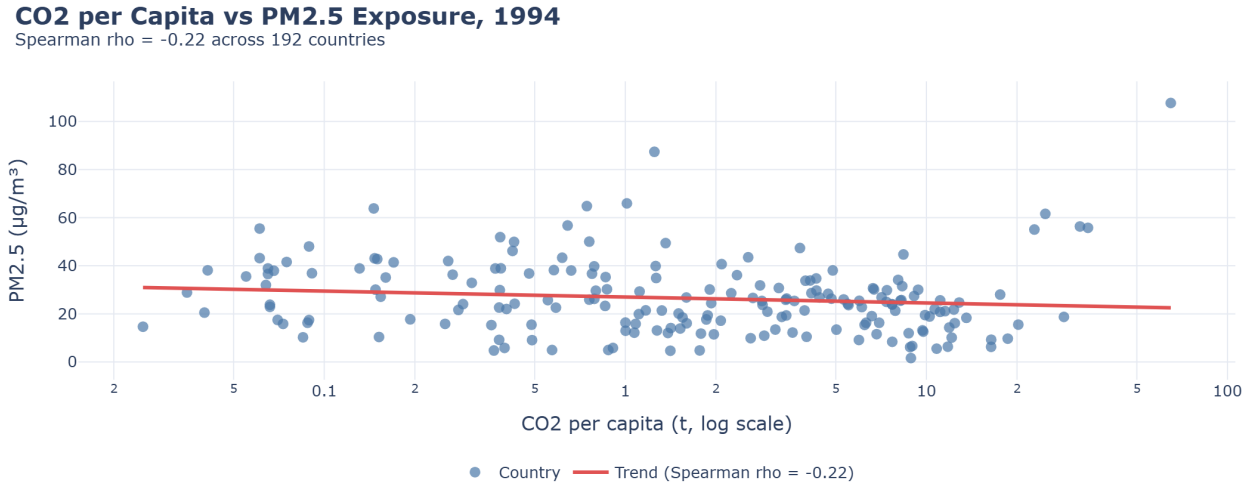
Q1. Which countries have the highest $PM_{2.5}$ exposure, and how has it changed since 1990?



SUMMARY

In 2023, the highest $PM_{2.5}$ exposure was in Qatar (108 µg/m³), Saudi Arabia (74 µg/m³), Bangladesh (69 µg/m³), several times the WHO guideline of 5 µg/m³. Since 1990, 4 of these 15 countries improved while 11 worsened, so high exposure is persistent and not uniformly improving.

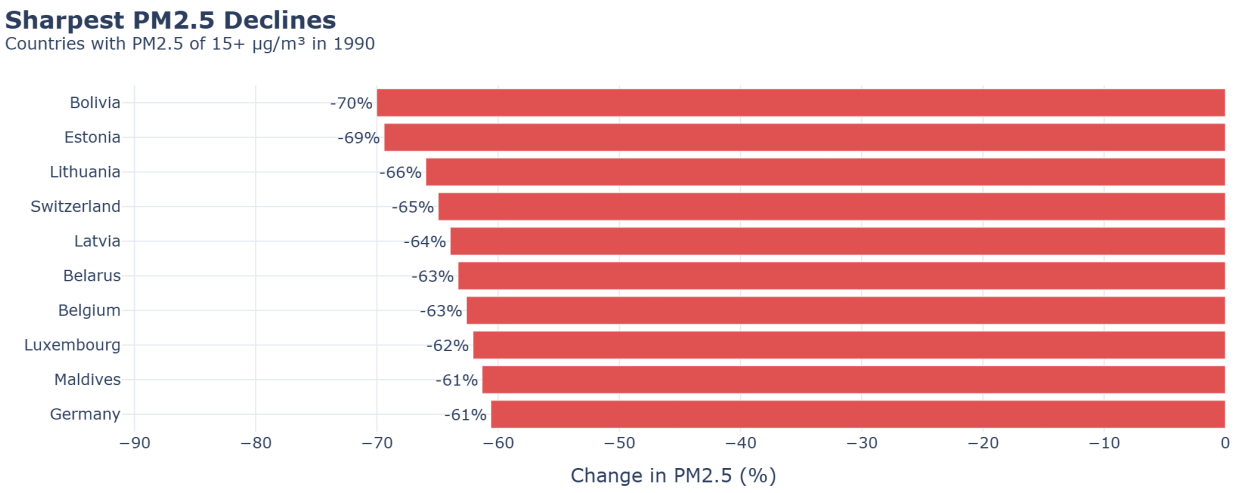
Q2. Do countries with higher CO_2 per capita have higher $PM_{2.5}$?



SUMMARY

Across 192 countries in 1994, the rank correlation is -0.22, i.e. a weak or negative relationship. High-income, high-CO_2 countries often have relatively clean air because of emission controls, while many low-emitting countries face high $PM_{2.5}$ from solid-fuel cooking, dust and industry. CO_2 per capita is therefore a poor predictor of local air quality.

Q3. Where $PM_{2.5}$ fell sharply, what changed (policy events)?

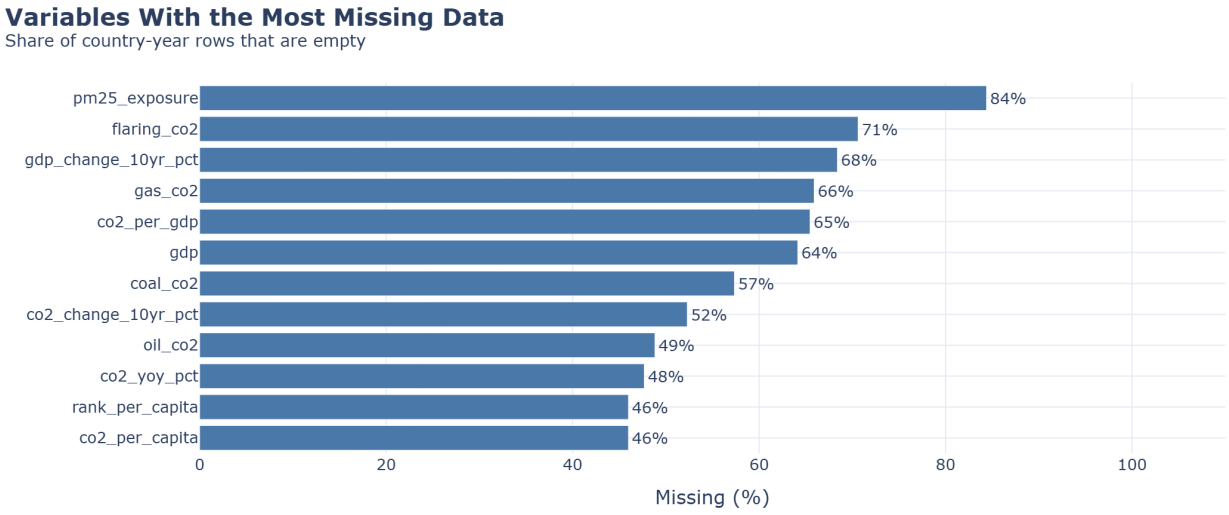


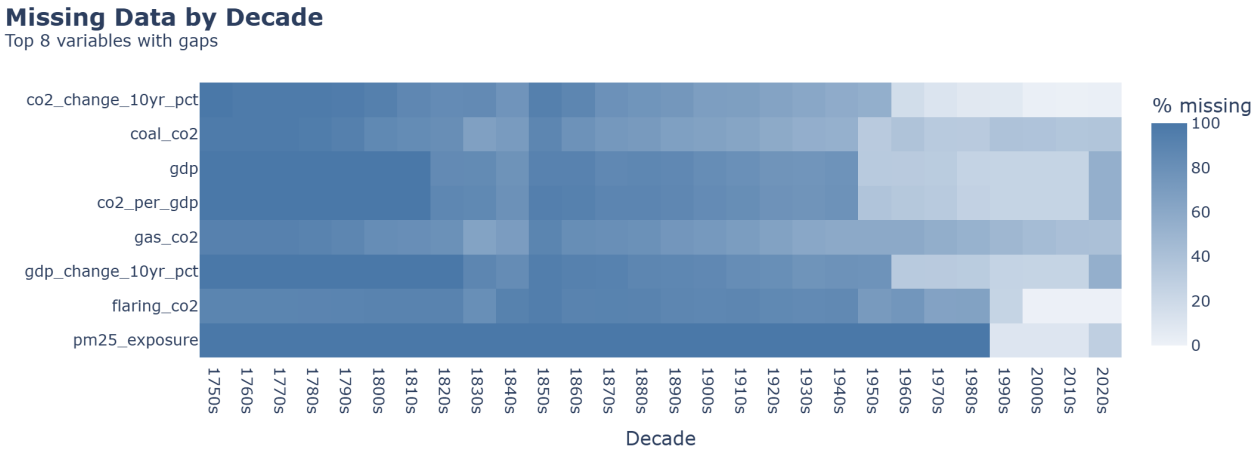
SUMMARY

The biggest declines since 1990 were in Bolivia (-70%), Estonia (-69%), Lithuania (-66%). Policy events recorded for these countries: EU Emissions Trading System launched (2005); Global financial crisis (2008-09) (2008); Kyoto Protocol enters into force (2005). Matching an event to a decline shows timing overlap, not proof; check each country's $PM_{2.5}$ curve around the event year.

Data Quality as a Finding

Q1. Which variables have the most missing data and in which years/countries?

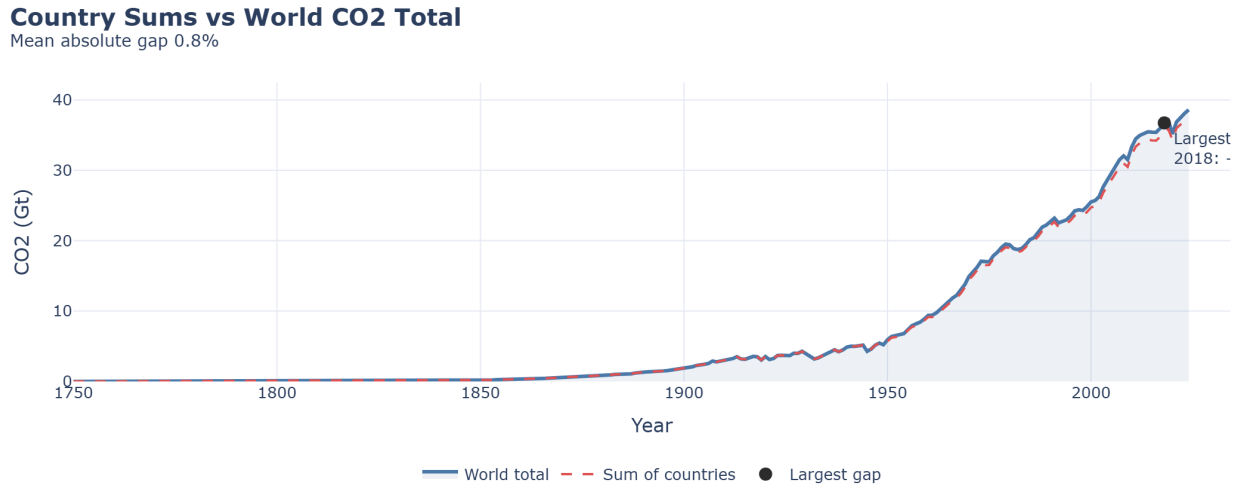




SUMMARY

The most incomplete variable is "pm25_exposure" (84% missing), followed by "flaring_co2" (71%) and "gdp_change_10yr_pct" (68%). Gaps concentrate in the earliest decades, and 23 countries have no "pm25_exposure" values at all. Any analysis using these variables is effectively limited to later years and a subset of countries.

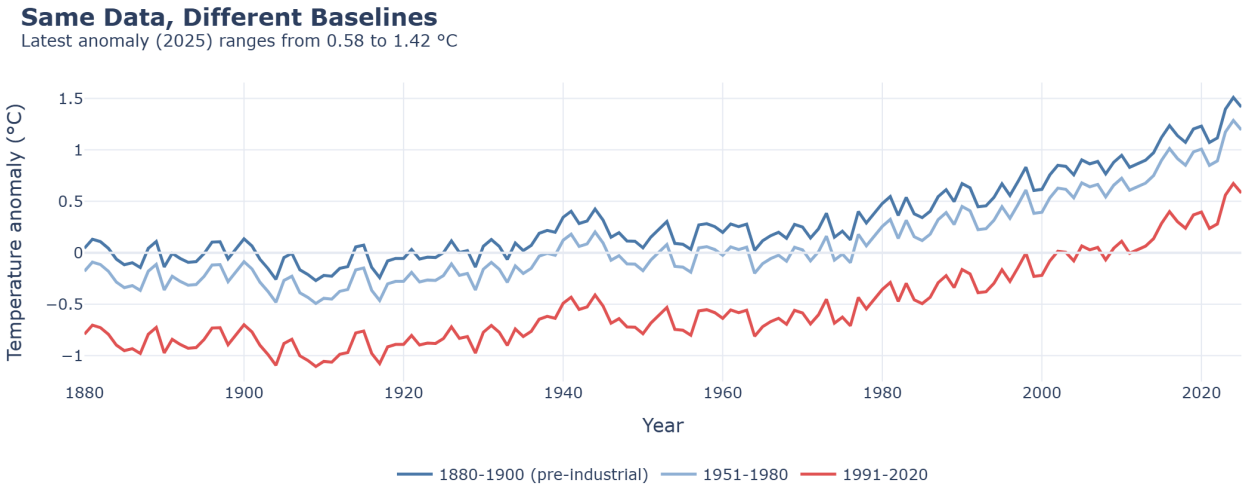
Q2. Do the country sums match the world totals?



SUMMARY

Summing countries gives, on average, a 0.8% difference from the reported world total, with the largest gap in 2018 (-3.4%). In 205 of 275 years the two agree within 2%. Differences typically come from international aviation and shipping, territories without ISO codes, and missing country-years.

Q3. How does the choice of baseline period change the temperature story?



SUMMARY

Rebasing the same temperature series changes the headline number: the 2025 anomaly is 1.42 °C vs 1880-1900 (pre-industrial), 1.19 °C vs 1951-1980, 0.58 °C vs 1991-2020. The shape and the warming trend are identical; only the zero line moves. A recent baseline makes warming look smaller and a pre-industrial baseline makes it look larger, so the baseline must always be stated when quoting an anomaly.